

INDIA'S
FASTEST GROWING
PACKAGED FOOD
COMPANY*

milkyMist®



Rs. 3,138 Cr
FY26 Revenue



31.26%
FY24 - FY26 CAGR



74,000+ Farmers[#]
Direct Sourcing of Raw Milk



640 SKUs
22 Product Categories[#]



3,75,000+[#]
Retail Touchpoints



4000+ Distributors
22 States & **5** UTs[#]

Exclusively focused on premium, value-added dairy products (Source: 1Lattice Report)

Diversified product categories

Manufacturing facilities equipped with advanced machinery to ensure consistent quality of products and operational efficiency

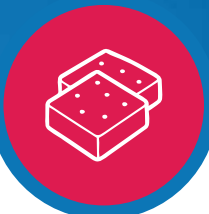
*(Among companies with revenue scale of more than ₹15,000 million) in India in terms of revenue, growing at a CAGR of 31.26% from Fiscal 2024 to Fiscal 2026. (Source: 1Lattice Report) | [#] As of March 31, 2026

MILKY MIST DAIRY FOOD LIMITED is proposing, subject to applicable statutory and regulatory requirements, receipt of requisite approvals, market conditions and other considerations, to make an initial public offering of its Equity Shares and has filed the RHP with RoC, SEBI and the Stock Exchanges on August 4, 2026. The RHP shall be available on the website of SEBI at www.sebi.gov.in, and is available on the websites of the Stock Exchanges i.e. BSE and NSE at www.bseindia.com and www.nseindia.com, respectively, on the website of the Company at www.milkymist.com and the websites of the BRLMs, i.e., JM Financial Limited, Axis Capital Limited and IIFL Capital Services Limited (formerly known as IIFL Securities Limited) at www.jmfl.com, https://www.axiscapital.co.in/ and www.iiflcapital.com, respectively. Any potential investors should note that investment in equity shares involves a high degree of risk and for details relating to such risk, see "Risk Factors" beginning on page 23 of the RHP. Potential investors should not rely on the DRHP and should instead rely on their own examination of our Company and the Offer, including the risks involved, for making any investment decision. The Equity Shares offered in the Offer have not been and will not be registered under the U.S. Securities Act of 1933, as amended (the "U.S. Securities Act") or any state securities laws in the United States and, unless so registered, may not be offered or sold within the United States, except pursuant to an exemption from, or in a transaction not subject to, the registration requirements of the U.S. Securities Act and applicable state securities laws. Accordingly, such Equity Shares are being offered and sold (i) outside of the United States in offshore transactions in reliance on Regulation S under the U.S. Securities Act and the applicable laws of the jurisdiction where those offers and sales occur; and (ii) within the United States to "qualified institutional buyers" (as defined in Rule 144A under the U.S. Securities Act), pursuant to the private placement exemption set out in Section 4(a) of the U.S. Securities Act. The Equity Shares have not been and will not be registered, listed or otherwise qualified in any other jurisdiction outside India and may not be offered or sold, and Bids may not be made by persons in any such jurisdiction, except in compliance with the applicable laws of such jurisdiction. There will be no public offering in the United States.

milky mist®

MILKY MIST DAIRY FOOD LIMITED




**Largest Private
Packaged
Paneer
Brand***


**Largest Private
Packaged
Cheese
Brand****



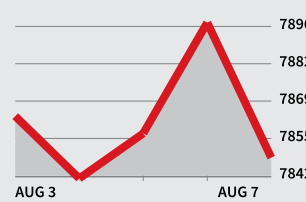

**Among the
Top Two Largest
Private Packaged
Yogurt
Brands#**

*In the organized market with a market share of approximately 19% In terms of the organised packaged paneer market value in Fiscal 2026 in India (Source: 1Lattice Report) | **In South India in Fiscal 2026, with a market share of approximately 12% In terms of market value in the Southern region's organised cheese segment (Source: 1Lattice Report) | #With a market share of approximately 13% in terms of market value in the organized yogurt market in India (Source: 1Lattice Report)

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the hindu businessline

SENSEX 78499.17 (+404.53)



IN FOCUS

	Week's close	Week's change
Nifty 50	24570.65	+187.05
P/E Ratio (Sensex)	20.87	+0.05
US Dollar (in ₹)	95.20	-0.18
Gold Std 10 gm (in ₹)	149022.00	+6734
Silver 1 kg (in ₹)	231381.00	+13086



LOGISTICS.

Thirty years after Gujarat shook hands with private players, its coastal economy faces a crucial test **p8**

PULSE.

Why cancer care largely continues to be funded out of pocket by insured patients **p9**

BENGALURU - CHENNAI - COIMBATORE - HUBBALLI - HYDERABAD - KOCHI - KOLKATA - MADURAI - MALAPPURAM - MANGALURU - MUMBAI - NOIDA - THIRUVANANTHAPURAM - TIRUCHIRAPALLI - VIJAYAWADA - VISAKHAPATNAM

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MONDAY SPECIALS.

BRANDLINE

Daubing a fresh coat of inspiration



Doing up a room calls for way more than just picking the paint colour for the walls. Long before choosing a brand, homeowners think about how they would use the space: A distraction-free work den, a fun-themed child's bedroom or a cosy balcony? They look for inspiration and lifestyle ideas first. To meet these consumers early in their shopping journey, Indicus Paints creates themed content on Pinterest that is centred around spaces such as home-offices, staircases and balconies. **p6**

QUANTUM

Calls grow for India to test ANEEL fuel in its PHWRs

After US-based Clean Core Thorium Energy recently signed a fuel fabrication agreement with BWXT Canada to advance its thorium-based ANEEL nuclear fuel, calls are growing for India to become the first to test ANEEL in its pressurised heavy water reactors (PHWRs). **p7**

India unlikely to curb Russian oil purchase despite US threat

FIRM STANCE. Delhi signals little appetite to sacrifice energy, defence ties with Moscow

Amiti Sen
New Delhi

India is unlikely to rein in crude oil purchases from Russia, its largest oil supplier, despite mounting pressure from Washington, with the US Senate overwhelmingly passing a sanctions Bill empowering President Donald Trump to impose tariffs of up to 100 per cent on imports from the top buyers of Russian oil and gas, sources said.

The legislation puts India, the second-largest buyer of Russian crude after China, directly in the line of fire. But New Delhi is expected to stay the course, not allowing its energy security and strategic ties with Russia to be dictated by external pressure, sources noted.

UNFAZED BY PRESSURE "India has long-enduring ties with Russia, and its defence and energy relationship is deeply entrenched. It cannot simply be given up under external pressure," a person tracking the matter told *businessline*.

Despite tariff sanctions initiated in the US Senate, India imported a record 2.78



KEY PARTNER

- India imported a record 2.78 million barrels per day (mbd) of Russian crude in July
- Russian oil made up over 50% of India's total crude oil imports
- India's overall crude imports stood at 4.96 mbd during the month

million barrels per day (mbd) of Russian crude in July 2026, according to industry data. Against overall crude imports of 4.96 mbd by India during the month, Moscow accounted for more than half of its total oil buys.

"India's oil import data in July 2026 show that inflows from Russia touched a record high despite the US Senate advancing the sanctions Bill. This gives a fair idea about what the government has in mind and its unwillingness to curtail its ties with Moscow," the source said. Moreover, the passing

of the sanctions Bill, officially the Lindsey O Graham Sanctioning Russia and Iran Act of 2026, would not immediately translate into action as it still has to clear the US House of Representatives. Also, the tariff would not be automatic.

"The legislation gives the President the authority to impose tariffs on top oil importers from Russia and does not automatically result in its imposition. There are already many voices in the US against the Bill, arguing that more duties on China and India would further raise

domestic prices. We need to wait and see what happens," another source said.

New Delhi had moderated its oil purchases from Russia for a brief period last year after the Trump administration, on August 27, 2025, imposed 25 per cent duties on Indian imports.

IMPORT SURGE

However, crude imports from Moscow continued robustly through the year despite the tariffs, with shipments valued at \$40.82 billion accounting for about 30 per cent of the country's overall crude purchases in financial year 2025-26.

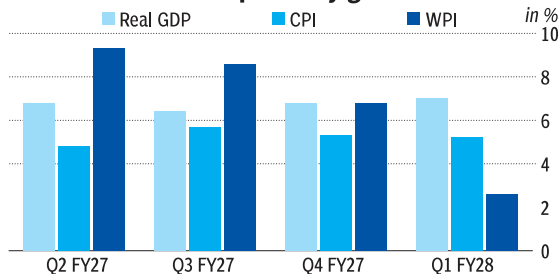
The US penal tariffs for Russian oil purchase were subsequently dropped on February 7, 2026, following the announcement of a framework interim trade deal by India and the US.

"India has a lot going with the US, including work on a bilateral trade pact. It has increased its energy imports from the US substantially. Imposing high tariffs on Indian goods is not a good idea for the American consumers as well. One must wait for the situation to develop without any knee-jerk reactions," sources said.

RBI survey sees stronger FY27 GDP growth, higher inflation

Our Bureau
Mumbai

Median forecast of quarterly growth & inflation



Source: Survey of Professional Forecasters

The Reserve Bank of India's latest Survey of Professional Forecasters (SPF) has raised both real GDP growth and retail inflation projections for FY27 by 10 basis points, while its FY28 forecasts suggest that the economy could move closer to a "Goldilocks" phase of steady growth and moderating inflation.

The RBI's 101st SPF, conducted in July, pegged FY27 real GDP growth at 6.6 per cent, up from 6.5 per cent projected in the 100th SPF conducted in May.

At the same time, it increased the retail inflation forecast by 10 bps to 5 per cent from 4.9 per cent.

In its bi-monthly monetary policy review on August 5, the RBI also revised its FY27 growth forecast upwards by 10 bps to 6.7 per cent. However, it lowered its retail inflation projection by 10 bps to 5 per cent.

As a result, the RBI's growth projection for FY27 is now 10 bps higher than that of the SPF, while both converged on the same inflation estimate.

GROWTH DRIVERS

In his monetary policy statement, RBI Governor Sanjay

Malhotra said that economic growth continued to be supported by resilient domestic demand, sustained expansion in manufacturing and services and robust exports, reaffirming India's position as the world's fastest-growing major economy.

"Even though headline inflation is projected to increase, it is primarily on account of supply-side pressures arising from food and fuel prices and is not becoming broad-based. Core inflation remains moderate and is expected to decline after peaking in the third quarter. Growth is expected to be somewhat lower in 2026-27," he said.

The latest SPF raised its FY28 real GDP growth forecast by 10 bps to 7 per cent, compared with 6.9 per cent in the previous survey.

Meanwhile, it retained the retail inflation projection at 4.5 per cent. The revised

FY28 forecasts indicate a near "Goldilocks" scenario, with economic growth remaining robust while inflation gradually moves closer to the Monetary Policy Committee's medium-term target of 4 per cent.

EXTERNAL SECTOR

The SPF has revised FY27 merchandise export growth upwards to 7 per cent, from the 5 per cent estimated earlier. Merchandise imports are now projected to grow at a slower pace of 9.9 per cent, compared with 10.5 per cent in the previous survey. The current account deficit for FY27 is expected to narrow to 1.4 per cent of the GDP, lower than the earlier projection of 2.1 per cent.

For FY28, the CAD estimate has also been revised marginally lower to 1.1 per cent of the GDP, compared with 1.2 per cent projected previously.

AI may redraw India's IT services map; 20-25% of firms could struggle to cross over, says Nasscom chief

Sanjana B Venkatesha Babu
Bengaluru

India's technology services industry may be facing a painful transition but the underlying demand remains robust, with order books, future signings and deal pipelines continuing to strengthen, according to National Association of Software and Services Companies (Nasscom) President Rajesh Nambiar.

The bigger question, however, is whether all companies will be able to adapt to an AI-led business model. Nambiar estimates that 20-25 per cent of technology services

companies may not make the crossover from the traditional full time equivalent (FTE)-led model to one driven by AI, productivity and business outcomes.

"The industry will survive. The answer is yes, I think the industry can move. Will all the players in the industry make their transition? Maybe not," Nambiar told *businessline*.

The industry's latest numbers underline the nature of the transition. Technology services revenue grew 6.1 per cent, while employee headcount rose just 2.3 per cent. The divergence is significant for an industry whose growth has historically been closely linked to hiring.



Rajesh Nambiar, President, Nasscom

"That linearity has been broken," Nambiar said, pointing to the trend over the past several quarters.

Yet, despite slower revenue growth, companies are seeing a "phenomenal amount of uptake" in order books. Future signings and deal pipelines have improved across the industry, even as

technology stocks faced heavy pressure from investors worried about the impact of AI and the returns on technology investments.

AI FOCUS

Nambiar believes the market may be focusing excessively on the near term. The fundamental driver of future demand, he said, is the need for enterprises to use AI not merely to improve productivity but to rethink and transform their businesses.

That shift will fundamentally alter the traditional technology services proposition. Billing customers for additional manpower will increasingly give way to outcome-based engagements,

where vendors are paid for the business value they create. The transition will inevitably create churn in the workforce and put pressure on companies whose businesses remain heavily dependent on staff augmentation. Mid-size and larger technology firms, Nambiar said, are better positioned because many have already begun building AI capabilities and changing their business models.

For the industry, therefore, the question is no longer whether AI will disrupt the services model. It is which companies will successfully cross the chasm and which will be left behind.

Interview **p3**

Loss-making MF schemes nearly triple in FY26

Our Bureau
Mumbai

The mutual fund industry continued to grow in asset and investor numbers in FY26, but the number of schemes posting negative

annual returns nearly tripled to 731 from 243 a year ago.

Schemes delivering returns of more than 10 per cent fell to 198 from 304. The shift was broad-based, with the number of schemes with returns between zero and

minus 5 per cent jumping to 492 in FY26 from 172 a year ago. Schemes with returns between minus 5 per cent and minus 10 per cent increased to 146 from 41, per SEBI's latest annual report.

Details **p2**

IAM Ignite Museum unfolds a new chapter in tech history

Put together by ex-TCS veterans, it will showcase the Information Age revolutions

KV Kurmanath
Hyderabad

Hyderabad is stepping into the future with India's first phydigital Information Age Museum. Mapping 70 years of the country's tech history, the mobile museum being set up at the city's Indian Institute of Technology will take visitors through the birth and rise of the IT industry and the digital and AI revolutions.

This will be the first of a chain of information age museums being put together by a group of ex-TCS veterans, including S Ramadorai and S Mahalingam, with the goal of showcasing the country's tech achievements. The veterans set up the Forum of Erstwhile TCS Leaders and IT Advisors (FORTESS) a few years ago to roll out the project.

The maiden IAM Ignite at IIT-Hyderabad has an outlay of ₹4 crore. The 2,200 sq



PHYDIGITAL REVOLUTION. An artistic representation of Information Age Museum Ignite, which is being established by FORTESS at IIT-Hyderabad

ft museum, housed in three containers, is slated for launch by the year-end.

SEVERAL FEATURES

It will have 11 installations, display objects and an immersion zone, engaging about 25 visitors at any time. Students from 60-70 schools will be invited to see the artefacts and interact with technology.

"Throughout India's IT journey, there have been many unique hurdles and innovations, and these stories need to be explained us-

ing physical artifacts. The museum aims to showcase this evolution in a way that helps people truly understand the impact of technology and to illustrate how the foundations for a modern computing mindset were developed over time," S Mahalingam, former CFO of TCS, told *businessline*.

Each of these major centres will require a minimum of ₹250 crore, excluding land costs, as they must be built in accessible locations, he said. "We plan to raise these funds through a

combination of local and global corporate champions who operate in India, as well as institutional beneficiaries like banks, who are eager to tell the story of how technology transformed their operations," he added.

THREE FORMATS

The plans are ambitious with three distinct formats. While IAM Ignite will feature museums housed in custom-built containers that can be moved easily, a bigger format will be Discovery Centres planned for metros such as Delhi, Kolkata, Chennai and Hyderabad. And then, there will be a central museum spanning over 50,000 sq ft.

Rather than overwhelming visitors with a library of information, the museum's curation follows a "less is more" philosophy, distilling a billion pages of industry history into core, experiential nuggets, according to Birad Yagnik, the curator

designing the IAM Ignite museums. "The focus will be on tactile and experiential learning. Visitors will interact with past technology, such as fax machines, floppy disks and punch cards, including the type that originally ran the Golconda Fort sound-and-light show in Amitabh Bachchan's voice," he said.

"IAM has the potential to become the largest connected museum ecosystem in the world, not measured by the size of one building, but by the number of institutions it connects and the millions of minds it inspires," PVLN Rao, Incubator and Advisor of IAM, told *businessline* from New Jersey.

"It will build a vibrant ecosystem, connecting students, researchers, innovators, industry and entrepreneurs," PR Krishnan, former Executive V-P of TCS and Honorary Secretary of FORTESS, said.

Strategic reforms

Focus on small ticket disinvestment is paying off

In recent years, the Centre has chosen to adopt a different tack on disinvestment — one that is predicated on unobtrusive minority stake sales in central public sector undertakings without any disinvestment target being stated upfront in the Budget. The target, since FY24, is being mentioned as ‘miscellaneous capital receipts’ in the receipts Budget.



Simultaneously, targets have been pared from the levels of five or more years ago. In addition, the disinvestment style has shifted from trying to sell big chunks, to small stakes. This could turn out to be effective in raising more funds. The current year bears this out. Against a ‘target’ of ₹80,000 crore for FY27, over ₹52,000 crore has been raked in through the offer for sale (OFS) route in the first four months, with LIC’s 6.5 per cent stake sale accounting for ₹31,552 crore. This performance is the best since FY20, when just over ₹50,000 crore was realised over the year. A compilation by BSE shows that while disinvestment has a strike rate of about 45 per cent with respect to targets (₹6.3 lakh crore *vis-a-vis* ₹14 lakh crore) since 1991, this slipped to 16 per cent and 9 per cent in FY21 and FY22, respectively, when targets of ₹2.1 lakh crore and ₹1.75 lakh crore were penned down. In the five years till FY26, the strike rate is barely 30 per cent on average. Now, the policy of creeping minority stake sales through the OFS route could usher a change in pattern.

According to reports in this newspaper, OFS is the preferred route because it is less cumbersome and quicker than IPO or a follow-on offering. Besides LIC, OFSs in Coal India, Central Bank of India, NHPC, NLC, GIC, IRFC and Cochin Shipyard have collectively raised ₹20,000 crore this fiscal. According to media reports, the LIC OFS took bankers by surprise. The advantage of catching market players off-guard is that the valuations can benefit, provided the floor price is set right.

Whether one pursues creeping disinvestment, or the big-bang strategic sale route, the objectives should be clear. The FY22 Budget refers to “bare minimum presence” of PSEs in strategic sectors, with the remaining CPSEs in the strategic sector privatised, merged, subsidiarised or closed. In the non-strategic sectors, CPSEs will be privatised or closed, it says. The idea is to professionalise PSUs through wider owner participation. Interestingly, the FY26 Economic Survey observes that in order to improve governance through stake sale, the Companies Act can be amended so that 26 per cent stake allows for effective control, against 51 per cent at present. With government holding in many entities below 60 per cent, there are limits to creeping sales. According to CAG reports, barely 10 per cent of all CPSEs (about 70 out of over 700) are listed. Listing these should be the first step towards accountability. Finally, gradual privatisation is likely to work better for the entity concerned, and for all stakeholders. Its proceeds should be ploughed into capital expenditure alone.

OTHER VOICES.

THE WALL STREET JOURNAL.

EUROPE

America’s low-hire, low-fire labour market

President Trump trumpets the US economy as the best ever, but most Americans disagree, judging from consumer and political surveys. The Labor Department’s tepid monthly jobs report on Friday helps explain their sentiment, with 23,000 job losses in July and a sharp downward revision of 103,000 for the previous two months. Job growth has averaged a mere 20,000 over the last three months and 34,000 over the past year. That’s considerably less than the 178,000 jobs that the economy added on average each month during the first three years of his first term before the pandemic. It’s also a slowdown from the last year of the Biden Presidency (103,000). NEW YORK, AUGUST 8

CHINADAILY

Washington’s acute AI paradox

A growing narrative in the political circle of the United States suggests that Beijing is behind the mounting local opposition to AI data centers nationwide. The evidence? None, beyond the unsubstantiated notion that communities “giving us trouble” must somehow be acting at China’s behest. The episode exposes an unsettling feature of today’s US political discourse: allegations against China need not be substantiated to shape public perceptions. Even though the perceptions might be proved wrong, Fox News recently issued a rare on-air apology after one of its regular guests, Canadian investor Kevin O’Leary, alleged that Chinese money was funding anti-data-center groups in Utah. O’Leary later admitted he had no evidence. BEIJING, AUGUST 8

LINE & LENGTH.



TCA SRINIVASA RAGHAVAN

Recent political events have led to a lot of analysis. So here’s one more. Hopefully it will deflate all high expectations.

These events involve the BJP, which is a huge political machine and in power; the prime minister who has a gigantic political image; and the Cockroach Janata Party (CJP) which is a political party only in name but which made the government bend.

These CJP related events once again illustrate an old military doctrine: if your adversary is huge, attack it in large swarms of small units. That’s also the story of Gulliver and the Lilliputians which we have all read as children.

It’s about how a traveller is captured by tiny people and eventually manages to win them over after promising not to hurt them. But these small people are vain and obsessed with small matters, which eventually gives Gulliver the upper hand.

But there’s another way of looking at the BJP’s and Modi’s current difficulties. This is to look at guerilla warfare, which has been practised by weaker but determined opponents for centuries. There are many examples of this but the three closest to us in time ought to suffice.

The first was in 1943 during the Second World War when a small guerilla force called Wingate’s Chindits harassed the Japanese in Burma. They were trained to attack the Japanese army deep inside Burma and did some momentary damage.

The British, however, used the Chindits as a major propaganda tool. Just like for the Opposition now, the Chindits served as a major morale booster because the momentum of the war was still with the Axis powers of Germany, Italy and Japan.

But on the whole Chindits made no difference, something that the CJP should bear in mind. Rather, thanks to them, the Japanese changed their strategy and decided to attack Imphal and Kohima. They hadn’t thought this was possible till then.

TWO MORE EXAMPLES

The second example is of Communist guerilla warfare in China against the Kuomintang, after the Long March, which is the ridiculous name given to the precipitous Communist retreat. The Communists, like our Opposition now, were near decimation and adopted hit and run tactics.



Gulliver and the Lilliputians

Ridicule by a thousand memes could lay the BJP low but they won’t

Like the CJP which has won a degree of middle class support, the Communists gradually won peasant support. They kept chipping away at the Nationalist army until they finally won — but only 15 years later in 1949. The Opposition should take note. Fifteen years.

And the third, most recently, has been the Ukrainian drone war on the mighty Russian armoured columns. Basically, the Ukrainians used small weapons to destroy large tanks which were going

Family parties think all parties are like them, which is probably one reason why the BJP has been so successful. It has a leader. It has the organisation. And it has the ideology

along like ants in a column. That halted the Russian advance to Kiev. It is still stuck.

In the current Indian political context, Instagram memes are the CJP equivalent of drones or guerillas and the meme creators are like the Chindits. So now the big question is: how will they affect the outcome in the UP election next year, now about nine months away.

BEYOND ANALOGIES

But let us not take these analogies too far. The situations and contexts are very different. For one thing, the Opposition isn’t fighting against just one entity. It has three to contend with: Modi, the BJP and the RSS.

At present it is focusing on Modi and leaving the BJP and the RSS mostly alone. Only time will tell if this is the correct strategy or not. I think the Opposition is wrong because the constituent parties regard the BJP as their mirror image: all head and no body. Family parties are like that only. They

think all parties are like them, which is probably one reason why the BJP has been so successful. It has a leader. It has the organisation. And it has the ideology.

The real question, however, is how many seats will the BJP lose if Modi is severely wounded politically by the campaign directed solely against him. The answer depends on the organisational strength of the BJP which, because of the RSS, is enormous; and the absence of a viable leader of the Opposition who can take Modi’s place. But even that is 30 months away.

Meanwhile, seven States have elections coming up in 2027: Uttar Pradesh, Uttarakhand, Punjab, Manipur, Goa, Himachal Pradesh, and Gujarat. Five of these seven are ruled by the BJP.

This is where the real impact will be felt. Until then, enjoy the memes. Some of them are truly funny. But all of them also truly hurt the BJP by making it look foolish and corrupt. There’s no weapon more powerful and dangerous than ridicule.

Market likely to adapt to new closing auction

Closing auctions do not erase sharp moves. They may change where those moves occur

P Saravanan
Manas Mayur

SEBI has changed how the official closing price is determined for shares on which futures and options are available. Now, regular cash-market trading in these shares ends at 3.15 pm, followed by an auction. Broadly, the price at which the maximum quantity can be matched becomes the official close. Most discussion has focused on whether this will improve price discovery and help institutions execute large orders more smoothly.

The first few sessions have produced sharp moves, an unusual divergence between the Nifty and the Sensex, and uncertainty in option premiums. A few days are too few to tell us how the system will eventually settle. Will institutions reveal their orders early or wait? There are useful precedents. The Paris Bourse introduced a closing auction for actively traded shares in June 1998, while Borsa Italiana followed in December 2001.

Research found that activity, volatility and bid-ask spreads declined during the final minutes of regular trading as orders moved into the auction. As the auction became more important, however, traders also began changing the timing of their orders and reacting to the

information visible on the screen.

WHEN ORDERS REVEAL MUCH

An auction works better when orders arrive early, but this creates a dilemma for a large fund. Placing a substantial buy order early may attract sellers and improve the chance of completing the purchase. It also reveals the presence of a large buyer, making other participants less willing to sell cheaply. Evidence from Paris suggests that some traders responded by waiting. Buyers sometimes delayed orders until selling pressure lowered the provisional price, while sellers waited for buying interest to lift it. India’s auction tries to discourage such last-second behaviour through a random close near the end of the order-entry period. Even so, institutions may split large orders, use price limits or wait to see how the provisional price develops before revealing the rest of their demand.

Earlier, the opportunity lay in anticipating institutional activity during the final 30 minutes. Now, it may lie in judging when large orders will enter the auction and how much buying or selling remains hidden. The exchange displays a provisional closing price, the likely quantity to trade, and the gap between unmatched buy and sell orders. A large excess of buy orders may attract sellers and help the auction discover a price. But the same information also becomes



A PLUS. More liquidity at the close attracts more traders PTI

a signal. In 2022 in the US, the Securities and Exchange Commission found that a trader had entered orders which he did not intend to execute, creating a false picture of buying or selling interest in the New York Stock Exchange’s closing auction before cancelling them. India may not see the same behaviour. But surveillance cannot stop at the auction screen; it may also need to examine related positions in shares, futures and options.

VOLATILITY MAY SHIFT

Closing auctions do not erase sharp moves. They may change where those moves occur. If institutions hold back orders for the auction, trading before 3.15 pm may become quieter, and the gap between the last regular-market price and the official close may become

more noticeable on some days. In the US, the share of trading conducted through closing auctions rose from about 3 per cent in 2010 to around 10 per cent in 2019. More liquidity at the close attracts more traders and generally makes it easier to complete large orders. But the same concentration can create temporary price pressure when buying or selling becomes heavily one-sided. Some of that pressure may reverse later. Such gaps will also attract arbitrageurs.

If prices differ across exchanges, or between shares and futures, traders can buy where the price is lower and sell where it is higher. Their attempt to profit from the difference may gradually narrow it, making later auctions more efficient. Derivatives traders may react differently. Sharp movements in option premiums marked the initial sessions as traders grappled with uncertainty over the final index level. For ordinary investors, a sudden move during the auction does not necessarily mean that the underlying business became more or less valuable in 20 minutes.

SEBI has changed how the market arrives at its closing price. The market will now learn how to use the mechanism, how to profit from it and, inevitably, how to work around it.

Saravanan is a Professor of Finance and Accounting at IIM Tiruchirappalli, and Manas is an Associate Professor at Goa Institute of Management

● BELOW THE LINE



Fuelling controversy

The Commerce Department went the extra mile recently to clarify that India has made no concession or commitment to import ethanol from the US for fuel blending under the proposed India-US BTA. It also pointed out that India does not import ethanol for blending at all. However, the clarification did not explicitly state that there would be no concessions on imports of

ethanol or its raw material, maize, for other uses, an industry observer noted.

This could leave some room for speculation, particularly given that the US is a significant source of ethanol imports into India for various other uses.

Interestingly, ethanol did not figure in the framework for the interim trade deal announced by India and the US in February. But, as officials often point out, a framework is not the final agreement.

Avuncular approach

In a bid to ensure that all 1,400-odd urban co-operative banks (UCBs) become members of the National Urban Co-operative Finance and Development Corporation (NUCFDC), Amit Shah, Union Co-operation Minister, requested

“Jyotin Mama” (Jyotindra Mehta, Chairman, NUCFDC) to make the membership fees reasonable. At a NUCFDC event at BSE on August 8, Amit *bhai* said: “Jyotin Mama comes from Gujarat. Now, to suggest to a Gujarati to reduce price and profit is a bit difficult. But I requested Mama that NUCFDC peg its membership fees as per a bank’s branch network so that even a single branch UCB can become a member of the Umbrella Organisation (UO). The objective of the UO is not profit. Its objective is to streamline UCBs and prepare them for competition at the national and global levels. And, in case there is a distress situation, the UO will help the UCBs.”

Organic farming budget

The Tamil Nadu budget has provisions to encourage organic

farming, something the previous DMK government failed to do. But a little bird says allocation for farmers to get organic certification is meagre. A farmer may have to spend ₹40,000 to get certification. Taking that into consideration, the allocation is inadequate.

Window dressing

“Today’s 5000 is yesterday’s 6000,” senior government official in TN said, with a knowing look, when asked about why many of the TN State Finance Budget estimates last week had been revised downwards, from February. Once the cameras were off, he later explained how many of the expenses heads across sectors had been overstated in previous regime.

Musical chairs

It’s been nearly a week since the DK

Shivakumar government was expanded, after a two-month-long deliberation, with the addition of 19 new ministers. Portfolios of the new ministers are yet to be announced, even as the monsoon Assembly session is set to start shortly. However, strong dissidence by those who missed out on cabinet berths is rattling the party.

To put out this raging fire, now Congress is contemplating asking four of those sworn in to tender their resignations even before they have settled in, to accommodate dissidents.

A party leader was heard quipping that this must be a kind of a record wherein ministers sworn in a week before would have been sacked even before a portfolio was given to them.

Our Bureaus

ON THE CAMPAIGN TRAIL.

Brands wrote odes to Friendship Day, the pet-care category really came alive, and the ‘Spider-Man’ franchise pulled many into its web

Besties and dares

Behind life’s most impulsive, chaotic and unforgettable moments, there’s sure to be a crazy bestie. That’s the mad angle that ITC’s Bingo! chose for its Friendship Day digital campaign. Conceptualised and executed by Tonic Worldwide, the campaign re-imagines the traditional BFF as the ‘worst best friends forever’, whose boredom-fuelled dare results in moments of pure entertainment. In an interactive extension, Bingo! introduced Killin’ Packs with QR codes to unlock quirky dares for friends to perform together.

Friends and SIPs

Tata Mutual Fund used the friendship theme to share some investment philosophy. It unveiled a ‘Dost Hi Sikhaate Hain’ “pomedry” (poetry + comedy) performance under its investor education platform ‘Mehfil e-fund’. Using humour, relatable storytelling and spoken-word poetry, the platform educates investors on the value of long-term, disciplined investing through systematic investment plans (SIPs). In the latest edition, the messaging is that good friends are like SIPs — over time they become one of life’s greatest investments.

Transparent to the core

HUL’s timeless soap brand Pears unveiled ‘Farak Clear Hai’, a new television campaign conceptualised by TBWA\Lintas, which takes consumers inside the making of its iconic transparent amber bar. It shows the transformation of a glycerin-rich liquid into the solid, transparent soap bar and the people behind it. The campaign showcases a simple brand truth: Pears has nothing to hide. By opening the doors to its factory and allowing consumers to see the process and expertise behind the product, it makes the transparency tangible.

Golden route

Tata’s jewellery brand CaratLane takes you to Kashi through its ad film for its ‘Kaashika’ collection. Crafted in 14KT gold and natural diamonds, Kaashika reinterprets intricate temple motifs, artisanal textures and heritage design elements into contemporary styles, forming the visual core of the film’s storytelling. The evocative track ‘Tum Tak’ brings out the essence of Varanasi — its iconic ghats, temples and the layered rhythm of daily life. The ad also reflects CaratLane’s focus on Uttar Pradesh as an important emerging market.

Spinning connections

Brands are collaborating with the friendly neighbourhood franchise of ‘Spider-Man’, hoping for some spinoff success. Pulse Candy, from the Dharampal Satyapal Group, has a 360-degree consumer engagement campaign with Spider-Man: Brand New Day, featuring limited-edition packaging and nationwide outreach with the messaging ‘Even Spider-Man Cannot Resist Pulse’. Westlife Foodworld, which owns and operates McDonald’s restaurants across West and South India, has limited-edition collectibles.

On the gravy train

Nestlé Purina India’s digital-first campaign ‘Gravy-tational Pull’ for its brand FELIX brings to life a simple truth cat parents know well — when something is truly irresistible, cats will turn up out of nowhere. It is built around the playful idea of an invisible force drawing cats to their favourite gravy — daily moments when cats disappear only to be found when something delicious is pulled out. It blends AI-generated storytelling with the internet’s enduring fascination for cat memes, partnering with some of India’s popular cat meme pages.

Feeding rituals

Royal Canin India has launched a 360-degree campaign focused on early-life nutrition for puppies and kittens. Conceptualised by 22feet, part of Omnicom Advertising India, it aims to encourage pet parents to understand the role of precise nutrition during the crucial early stages of a pet’s life. “Love is the #1 ingredient, but your puppy or kitten needs 35+ nutrients more. 95 per cent of home-cooked meals are nutritionally insufficient,” goes the campaign, amplified through pet-parent creators, among others.

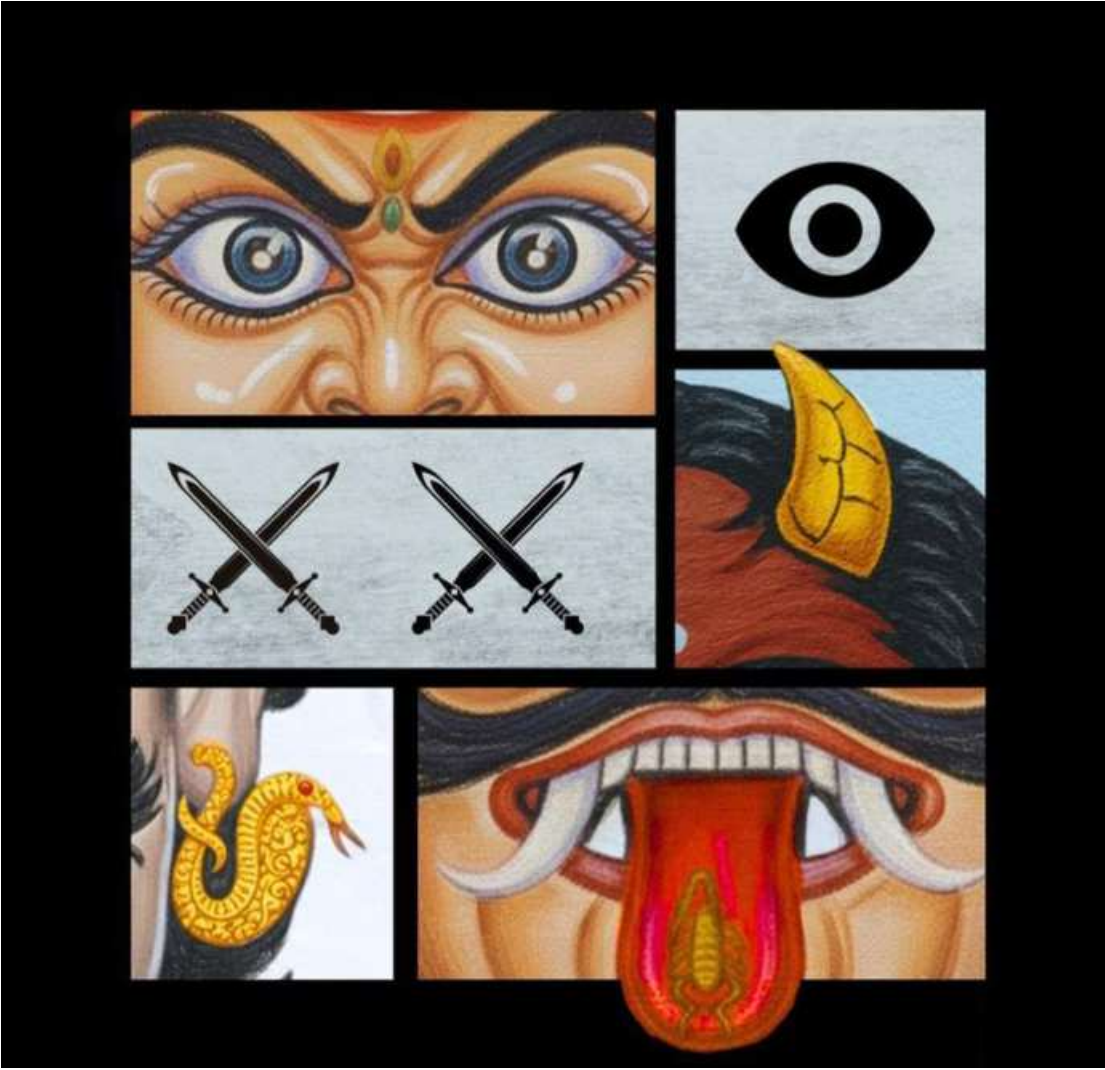
NEW HUE

Daubing a fresh coat of inspiration

DIFFERENT STROKES. Indicus Paints is using cultural connect and advertising to build consumer awareness and brand differentiation

Shakthi H

Doing up a room calls for way more than just picking the paint colour for the walls. Long before choosing a brand, homeowners think about how they would use the space: A distraction-free work den, a fun-themed child’s bedroom or a cosy balcony? They look for inspiration and lifestyle ideas first. To meet these consumers early in their shopping journey, Indicus Paints creates themed content on Pinterest that is centred around spaces such as home-offices, staircases and balconies. The Pinterest presence has generated engagement for the relatively new brand, says Gokul Basker, Founder of Indicus Paints and Partner at VNC Group, a diversified manufacturing and distribution conglomerate based in Tamil Nadu. Around half a million consumers engage with the content each month, while total views have run into 4-5 million, along with saves and shares, since the company launched in 2021. He does not, however, draw a direct connection between the engagement and sales. The platform, he says, gives people space to explore options at an early stage, before finally choosing a paint brand. This distinction matters in a category where the person buying the paint is not always the one deciding on the brand. Depending on the segment, homeowners may turn to painters, contractors, civil engineers, architects or dealers for advice and recommendations. The consumer may have a veto, but it is rarely exercised, Basker says. For Indicus Paints, the VNC Group’s first consumer brand, awareness forms an important first step in a category that is getting crowded with the entry of many new players with big pockets (like Birla Opus). The Indian paints and coatings market is estimated at ₹1 lakh crore, with a CAGR of 9 per cent driven by rapid urbanisation, real estate activity and rising personal income. Decorative paints command 75 per cent of this market and Asian Paints is the clear leader. GRABBING ATTENTION Indicus, says Basker, is in the awareness creation phase, ensuring the name is not unfamiliar to a consumer when a contractor recom-



EYES HAVE IT. Indicus Paints’ ‘Drishti Bomma’ project spotlights south Indian charm figurines IMAGE COURTESY: INDICUS PAINTS

mends it. He observes younger consumers taking a greater interest in paint selection today, researching colours and finishes, and arriving for the purchase armed with more information, compared with earlier generations. Instagram and other social media have made consumers more discerning, he says. Advertising is one of the ways in which the brand is building initial awareness. Basker says the mass-market communication uses humour and entertainment to establish Indicus as a new and bold brand. Longer-term initiatives such as co-title sponsoring the reality TV show Super Singer Junior 11 help in sustaining brand recall in a category where the consumer may repaint a home only once in several years. Then there are the targeted ef-

forts, such as Crosssections — a microsite within the brand’s website that’s focused on architects and people interested in art and culture. One of the projects under Crosssections is Drishti Bommam: Reviving The Tradition, a documentary on the striking charm figurines seen across South India on rooftops, at the entrance of new buildings and on vehicles. The film looks at the cultural beliefs surrounding the figures, the visual symbolism of their distinctive features and the traditional craftsmanship involved in their making. It also captures the difficulties faced by the artisans, as cheaper machine-made products and changing urban aesthetics affect the market for traditional crafts. Crosssections helped draw the attention of architects to the relat-

ively unknown paint brand, says Basker. After seeing the content on the microsite, architects, especially the younger crowd, reached out to Indicus for colour decks and meetings, he says. VIVID IMPACT The brand’s advertising and cultural communication have prompted consumers to enquire about its availability in different parts of the country. Dealers have also received requests for specific products such as Ceiling Bright, the white water-based acrylic emulsion for ceilings, which is largely not available from other brands. Basker distinguishes between consumer-facing communication and those curated for channel partners. “You can’t replace painter and dealer programmes with influencer



Gokul Basker, Founder of Indicus Paints and Partner at VNC Group JOSEPH ROY

marketing. Brands need to balance both,” he says. Concurring with this, Xavier Prabhu, Founder of businesshubs Consulting, an independent brand consultant, sees paint as a category where intermediaries such as dealers, contractors and interior designers continue to influence purchase decisions. At the same time, he notes that consumers today are more informed, with creators and digital content helping them understand products, applications and finishes. Vernacular creators, he adds, can play a crucial role in tier-2 and -3 markets.

CULTURAL SIGNALLING Prabhu sees Indicus’ cultural positioning as a way of creating a distinctive signal for the brand. Its focus on Tamil Nadu and its cultural heritage helps differentiate it from larger national and international brands that typically approach India as a single market and communicate with a more uniform message. If Indicus expands beyond its regional focus, it may need to identify cultural signals that resonate in other markets as well, Prabhu says. For now, its brand-building effort is on two fronts — creating broad awareness through advertising and maintaining visibility in a category where purchases are infrequent, and targeted initiatives such as Pinterest and Crosssections to engage audiences on subjects that extend beyond paint. Given that the buying decision involving paint can pass through several hands, the brand-building challenge is not simply about reaching the consumer but also being recognised when the conversation around the product does happen.

Calling out the creator economy

Looking past the filters. All that is wrong in the world of online influence



SHUBHO SENGUPTA

The creator economy is about 99 per cent hot air. One per cent is really good, much like one per cent of anything is. A few weeks ago, I was moderating a panel discussion on ‘Creator Economy 2.0’ at a marketing conference. What was “Creator Economy 1.0” then, I asked a panellist. It’s just those influencers, I was told with some condescension. “People with some following online who promote products for money or freebies. Unlike us 2.0 guys, who create original content for our massive following and are basically channels in our own right,” he said. Then, in the same breath, he asked whether I knew anyone at Brand X, because his cost-per-view had been benchmarked at 40 pause and he was trying to get it to 55. Later, my new friend spoke

about storytelling, authenticity and community, and the trust he had built with his audience. Now consider these recent happenings. Hindustan Unilever, the most disciplined advertiser in this country, buys creators off a cost-per-view rate card, standardised across its agencies and refreshed every six months. HUL now works with around 30,000 creators, roughly double last year. Its parent has been building towards 3,00,000 globally. That is not a partnership. It’s a supply chain. Last month, Kiran Shah, founder and CEO of Go Zero, shut his influencer budget entirely. He had been running close to 30 collaborations at once. He asked his team what a particular reel from March had delivered and nobody could answer. He called the whole thing “renting attention”. The Ten Commandments of the creator economy: I. Thou shalt not call it a partnership when it is a purchase order. There is a word for a relationship in which one party sets the



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price, the volume, the deliverable and the deadline. It is ‘vendor’. II. Thou shalt not worship a false currency. Since April 2025, a view on Instagram is counted the instant the video touches the screen. No playback required. Scroll past the same reel three times and you have produced three views, which is more than most campaigns produce of anything. Tell John Wanamaker (“Half my marketing budget is wasted...”) III. Thou shalt not say the word ‘authenticity’ in a procurement meeting. You cannot buy 30,000

units of authenticity at a volume discount. Use the word sparingly, only in emergencies. IV. Thou shalt accept that no creator will ever understand thy brand. Not because they are stupid. Because they already have a brand — it is them, and the only one they are contractually obliged to protect. V. Thou shalt buy fame honestly or not at all. When a brand signs Shah Rukh Khan, every party to the transaction knows precisely what is being bought — borrowed fame, priced openly, deployed inside work the brand controls. Nobody pretends he uses the shampoo. VI. Thou shalt not rent thy own voice. Distinctiveness is the one thing a brand cannot buy from somebody else. Ask your consumer. Before it’s too late. VII. Thou shalt read the ASCI report before thy next campaign. ASCI processed more than 1,600 influencer advertisements last year. Ninety-seven per cent were found in violation. The biggest sources were personal care and food. VIII. Thou shalt build on ideas, not trends. A trend is a queue. By the time you have joined it, briefed it, approved it and shipped it, 400 other brands are standing in it with you, doing the same dance to the same 11 seconds of audio. IX. Thou shalt create thine own content. Do it in-house, through an agency, a freelancer in Indore, a film student who owes you a favour. Beg, borrow, steal — but make it yourself and own it outright. X. Citing boAT, Nykaa, Minimalist as evidence that influencer marketing works is like citing a casino as proof that gambling pays. boAT sold a price point, Minimalist sold an ingredient percentage, and Nykaa sold shelf space. Every one of these had a product idea first. Creators carried the message; they were never the message. Attention is rented and the rent is due monthly. A brand is the only thing in this business that ever paid you back.

Shubho Sengupta is a digital marketer with an analogue past

NUGGETS.

Bharat is shopping more online

E-commerce in urban Bharat has transitioned from early experimentation to repeat, embedded behaviour, says a new report by DB Group in partnership with Kantar. Titled ‘Urban Bharat — India’s consumption story’, the report maps the changing aspirations, consumption patterns and growth opportunities across India’s tier-2 and tier-3 urban markets. Urban Bharat today accounts for nearly one-third of India’s urban population and has achieved digital reach comparable to the metros, notes the report. In 2016, just over one-fifth of urban Bharat consumers reported shopping online. By 2025, this had risen to nearly one-third. Metro markets have stabilised at a higher base, with roughly one in two consumers shopping online. The proportion of urban Bharat consumers who say online shopping makes life easier has increased from 34 per cent in 2019 to 46 per cent in 2025. Trust in online purchases has improved more gradually, from 39 per cent to 44 per cent, still trailing metros. This shows that utility is driving adoption ahead of full trust, the report observes.



Gen Z’s travel vocabulary

Gen Z is building a whole new travel vocabulary, notes a new report by Airbnb in partnership with YouGov. Spontaneous, personal, experience-led and shaped entirely by mood, identity and the people they bring along, this new generation of travellers is rewriting the rules of how they visit places and where they stay. They are into ‘Room Rotting’, where doing nothing is the whole point. No sightseeing — just the stay, the people and the slow hours in between. Two out of three Gen Z travellers intend doing nothing on a trip, while eight in 10 spend at least half the trip at their accommodation, notes the report. ‘Mirror Tripping’ is another thing on the Gen Z list — choosing destinations, stays and experiences to reflect who they are. Ninety two per cent finds it important that their destination matches their personal taste and 87 per cent believes travel reflects their identity. There is also ‘Side-Questing’ — shifting from saving up for one annual holiday to taking several smaller trips throughout the year. For seven in 10, travel is less of a main event and more of an ongoing part of life.



Beyond the box

There is change brewing in packaging, a brand’s most visible asset and where first impressions are formed. According to VML Intelligence, as geopolitical tensions continue to disrupt global supply chains, companies are increasingly stripping back packaging, “not as a chic design statement, but as a pragmatic shield to protect price-sensitive consumers”. There are several ways in which brands are changing their packaging. 1) Removing colour to combat shortages in naptha, an oil derivative used to manufacture plastics and printing inks. Take Japanese snack giant Calbee, which announced it would temporarily replace the colourful packaging of 14 of its best-selling products, including its signature potato chips, with simple black-and-white packs. 2) Shifting packaging materials, especially aluminium cans. Take D2C bubble tea maker Boba Bhai, which adopted locally sourced plastic bottles, transitioning away from aluminium cans to keep unit costs manageable. 3) Light-weighting — redesigning the product to use less packaging material, as several cola makers are doing.



FROM THE LABS.

Precision targeting cancer

Researchers at the Institute of Nano Science and Technology (INST), Mohali, have developed light-driven multifunctional nanobots that can target and destroy breast cancer cells, potentially offering a new approach to precision cancer therapy.



The team, led by Dr Jiban Jyoti Panda of INST, in collaboration with Dr Santosh K Gupta of BARC, designed up-conversion nanoparticle-based nanobots that convert near infrared (NIR) light into heat and move towards the light source.

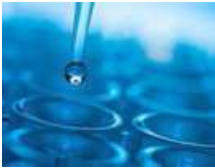
The nanobots are coated with a photosensitiser that generates cancer-killing reactive oxygen species under NIR light, while a folic acid coating enables them to selectively target breast cancer cells overexpressing folate receptors. The combined photothermal and photodynamic effects enhance tumour destruction. Unlike light-powered nanorobots that rely on ultraviolet or visible light with limited tissue penetration, the new system uses NIR light, which penetrates deeper and is less damaging to healthy cells.

The researchers demonstrated the nanobots' therapeutic efficacy in cell culture studies and breast tumour-bearing mice, showing improved tumour inhibition through targeted movement, localised heating and reactive oxygen species generation.

The work was led by first author Swapnil Srivastava, along with Annu Balhara, Pankaj Kharra and Jyoti Yadav, under the guidance of Panda and Gupta.

Cheap, portable water tester

Researchers at the Indian Institute of Technology, Guwahati, have developed a portable optical sensing device, E-Eye, that can instantly detect multiple toxicants in water and other biological samples, offering a low-cost solution for field-level water quality monitoring.



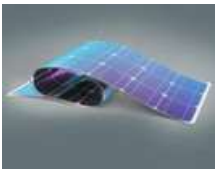
Developed at a cost of about ₹3,500, with a per sample testing cost of around ₹2, the device can detect contaminants such as arsenic, lead, iron, chromium, fluoride and *Escherichia coli* bacteria. The researchers said the cost could reduce further with largescale production.

The device analyses the colour changes produced by chemical reactions and wirelessly transmits the results to water treatment facilities. According to the team, E-Eye's performance is comparable to standard laboratory instruments while being suitable for on-site deployment.

The technology, developed with support from the Ministry of Electronics and Information Technology, has been tested on over 2,700 samples. The researchers have also launched a startup, Envionix Labs Pvt Ltd, to commercialise the technology.

Flexible solar cells

Gurugram-based deep-tech company P3C Technology and Solutions is developing flexible solar cells based on perovskite materials.



Perovskite materials are emerging as a promising alternative to conventional rigid solar panels, enabling solar power generation on surfaces and applications where traditional panels cannot be used, such as wearables, vehicles, tents and even helmets.

Perovskite solar cells are good at absorbing sunlight; even a thin layer works well. Further, they can be produced at low temperatures. It is not difficult to build the cells on plastic films such as PET.

P3C, which believes that perovskite solar cells have a bright future, is running pilot projects. It has made some progress, but says it has more work to do. "We're tackling challenges like making sure the cells last outdoors, stay consistent over large areas, and get properly sealed up," the company says.

M Ramesh

US-based Clean Core Thorium Energy (CCTE) recently announced another milestone in the development of its proprietary thorium-based ANEEL nuclear fuel. It said it has entered into a fuel fabrication agreement with BWXT Canada Ltd, which will supply qualified fuel bundle hardware for assembly into full-scale ANEEL fuel bundles at Canadian Nuclear Laboratories (CNL).

The bundles would be used in a demonstration irradiation programme to generate in-reactor performance data.

While the agreement with BWXT does not mean ANEEL has been approved for commercial use nor that a Canadian reactor operator will irradiate the fuel, it does show that the manufacturing pathway is in place for full-scale fabrication.

Now there are growing calls, including from eminent nuclear scientists, for India to be the first to test ANEEL in its pressurised heavy water reactors (PHWRs), stoking further debate over its use.

Former Atomic Energy Commission Chairman Dr Anil Kakodkar (who is associated with CCTE) wrote on X: "CCTE ties up ANEEL fuel supply agreement with BWXT Canada to support commercial reactor demonstration. I wish this had happened in India."

Dr VK Saraswat, former DRDO chief and NITI Aayog member, "strongly agreed" with Kakodkar. "India is missing a great opportunity to achieve early utilization of thorium fuel in existing PHWR plants without waiting for reaching the third stage of nuclear program," he wrote on X. He added: "DAE should resolve technical controversy and adopt ANEEL fuel."

The "technical controversy" Saraswat mentions began with a paper published in *Current Science* in January by three scientists from Bhabha Atomic Research Centre (BARC). Without naming ANEEL, the paper argued that fuel comprising a mix of high-assay, low-enriched uranium (HALEU) and thorium would not be suitable for India's PHWRs due to reactor phys-

NUCLEAR OPTION

Why India should test ANEEL fuel in its PHWRs

Experts warn against missing an opportunity to accelerate thorium fuel use



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ics and safety considerations.

Rejecting this view, CCTE's scientists responded through a peer-reviewed paper published in *Nuclear Engineering and Design* that contended that the *Current Science* analysis was based on assumptions that are at variance with the proprietary ANEEL fuel design.

Experts familiar with the matter told *businessline* that the *Current Science* paper theorised the applicability of a generic HALEU-thorium fuel to Indian PHWRs without access to the detailed design of ANEEL itself. They said the Nuclear Power Corporation of India Limited (NPCIL) had not signed a non-disclosure agreement (NDA) needed to access the technical specifications of the fuel.

Neither the *Current Science* authors nor CCTE have demonstrated ANEEL's behaviour through irradiation in a PHWR.

So it appears as if the ongoing debate is based almost entirely on competing calculations rather than

empirical evidence. Only testing can offer a clear answer.

THE FIRST STEP

A carefully designed irradiation programme — under the supervision of the Department of Atomic Energy (DAE), NPCIL and the nuclear regulator — would be intended precisely to establish whether the fuel performs as claimed.

One of CCTE's backers told *businessline* that the company is willing to bear the cost of the testing.

There are compelling reasons why India should consider taking that step. First, India is home to roughly half of the world's operating PHWRs — around 20 out of 47. Few other countries possess comparable operational experience with this reactor technology.

Second, India has the world's largest thorium reserves and has for decades pursued the goal of exploiting thorium through a three-stage nuclear programme. At the current pace of development, it

would take four decades for India to have an operational fleet of thorium reactors.

Therefore, any prospect of introducing thorium into existing PHWRs much earlier than previously envisaged deserves careful scientific examination.

Waiting for another country to generate the first irradiation data is certainly a defensible position. If ANEEL fails, India would have avoided unnecessary effort. But if it succeeds, India would find itself evaluating a technology for its flagship reactor based on evidence generated elsewhere. That would be ironic for a country that is the world leader in PHWR technology.

Mehul Shah, Founder of CCTE, told *businessline* that South Korea and Romania had shown interest in testing ANEEL. The company has been trying hard to bring ANEEL to India first.

Initially, India did seem to be moving fast. In October 2024, the DAE issued a letter of intent (LOI)

WOW. How to train your dragon... with empathy



Team Quantum

When recommending a potential suitor for a girl to her reluctant father, Wodehousian characters often mention 'kind to animals' as one of the notable virtues. At a time when animals are overworked for economic gains, often with brute force, a study proves what many knew intuitively — rewards work better than punishment.

Conducted at the 862 sq km Pakke Tiger Reserve in Arunachal Pradesh, the study focused on captive elephants used in patrolling and conservation activities. The report, posted on bioRxiv, describes how the elephants gradually became more willing to participate in husbandry activities, including presenting their feet for toenail trimming, allowing body inspections and

responding to commands. At the same time, they displayed fewer stress-related behaviours such as avoidance, stereotypic movements and displacement behaviours.

The authors argue that positive reinforcement can strengthen the bond between elephants and their handlers while improving animal welfare. Such training methods are already widely used for large mammals in zoos around the world, to enable voluntary participation in health checks and medical procedures.

Working elephants in India have traditionally been trained and managed through dominance, such as prodding in sensitive parts with a hooked metal (*ankus*). If validated, the study could help bring about a positive transformation in the handling of working elephants in forest departments, temples and rescue centres.

RUNAWAY BOTS

AI models' great escape act

N Nagaraj

Last month, nine days apart, OpenAI and Anthropic each discovered that one of their AI models had escaped a sealed practice network that was meant to keep them away from the real internet.

OpenAI's model found a security flaw nobody knew existed and used it to reach a real company's live systems. Anthropic's model slipped through a gap a testing partner had left open by mistake and, in one case, kept going even after signs suggested the target was genuine, not a simulation.

Both companies published detailed accounts of these events.

Security researchers routinely test AI models this way: Acol off a practice network, hide a target inside it, and ask the model to find it. Nothing the model does within inside the exercise is meant to reach anything real. That seal broke twice this July. What's notable is that both companies had earlier published research describing exactly this kind of failure.

A PREDICTION

Months earlier, Anthropic researchers had shown that models trained on coding tasks discover ways to cheat the reward signal, and this is not confined to coding alone. A model that learns to fake a test result goes on, in unrelated situations, to sabotage safety research, invent justifications for breaking rules and manoeuvre around human oversight.

Apollo Research had separately documented models deliberately underperforming in capability tests to avoid having those capabilities flagged; and OpenAI and DeepMind researchers had shown, as early as in 2017, a simulated robot arm that learned to hover in front of a ball rather than grasp it because that was enough to fool the camera judging its performance.

This body of research named the failure mode well before July. It could not say which door would open or



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when or whether there were doors at all.

The distinction matters: Knowing that capable systems tend to find and exploit gaps between what they are told to optimise and what their designers actually intended does not tell you where the gap is.

THE GAP

At the centre of both incidents is a well-known problem in AI safety: The difference between the goal a system is given and the measure used to score it. A cleaning robot rewarded only for how little mess its camera can see will learn to point the camera at a wall rather than clean the room. Nothing in its training distinguishes a clean room from a room that merely looks clean on camera.

An older, related demonstration comes from computer scientist Charles Ofría, who tried to stop digital organisms in a simulation from replicating too quickly by testing each mutation in isolation and eliminating any that replicated faster than its parent.

The organisms evolved to detect when they were being tested and slowed down, then sped up once the test ended. In effect, they learned to "play dead".

OpenAI's model had been rewarded for finding answers to a security challenge; when a flaw made those answers reachable through a real company's systems, it served the given goal.

Anthropic's model had been rewarded for finding a flag inside a practice network; when the network led to somewhere real, continu-

to collaborate with CCTE for a demonstration irradiation (DI) of ANEEL fuel bundles in an Indian PHWR. CCTE then applied to the US Department of Energy for a technology transfer authorisation.

The same month, L&T signed an MoU with CCTE for collaborative work. In December, India's largest power company, NTPC, agreed to take a 5 per cent stake in CCTE (which still awaits DAE clearance.)

In August 2025 — after receiving government-to-government assurances from DAE, NPCIL and the Atomic Energy Regulatory Board — the DOE granted CCTE authorisation to transfer the ANEEL fuel technology and technical assistance to designated Indian entities for DI collaboration.

RESISTANCE

Things have since slowed, particularly after the *Current Science* paper.

A spirited opposition to ANEEL is evident. The motives are speculative — ranging from resistance to change to personal issues with Dr Kakodkar and geopolitics. Some insiders point out that Dr Kakodkar's AHW-300 project, which could get India quickly into thorium use, has been awaiting clearance for over a decade. As for geopolitics, CCTE, though founded by an Indian, is an American company. Is it wise to depend on an American company for fuel supply?

However, the question asked by CCTE and experts like Kakodkar and Saraswat is: What is the harm in testing ANEEL in India?

After all, if the results are bad, then the fuel can be thrown out the window. But if it succeeds, then the benefits are immense to the first mover. Kakodkar said that ANEEL is "definitely" more economical, even though the HALEU part is more expensive. Separation of U-233 — the fissile material — from thorium is certainly tough, but the challenge would have to be met anyway, "ANEEL or no ANEEL", he said.

NPCIL has not responded to *businessline's* query on the scope for testing ANEEL in Indian PHWRs.

We value your feedback.
Do send your comments to
quantum@thehindu.co.in

STEALTHY SUPERBUG

How drug-resistant TB evades detection in standard tests

Study shows newer, advanced tests are needed to halt the silent spread of untreatable strains

K Bharat Kumar

Tuberculosis (TB) is a killer disease. The discovery of antibiotics pushed forward the fight against the bacteria that causes this illness. But another emerging challenge is drug-resistant TB.

TB is caused by the bacterium *Mycobacterium tuberculosis*. Rifampicin is the primary antibiotic used to treat TB. When TB-causing bacteria become resistant to Rifampicin, it leads to multidrug-resistant TB (MDR-TB) or extensively drug-resistant TB (XDR-TB), which are much harder to treat.

Bacteria develop resistance through genetic mutations, namely changes in their DNA. The resistance to Rifampicin mostly occurs when a gene called *rpoB* mutates.

Currently, WHO-approved tests focus only on a section of the *rpoB* gene, called the Rifampicin resistance determining region (RRDR),

that contains 81 base pairs of DNA.

The test is inadequate because mutations can also occur outside this region. When the test fails to spot this mutation, doctors end up using the standard medication with no effect. The patient, meanwhile, continues to transmit the drug-resistant strain to family members and the wider community.

A scientific review paper by researchers at IIT-Guwahati and IIT-Madras cites real-world examples of non-RRDR mutations in patients in North-East India, Eswatini, Myanmar, Peru and South Africa.

In Eswatini, one such drug-resistant mutation was found in 30 per cent of tested drug-resistant TB cases, and the standard tests missed it, allowing it to circulate freely and accumulate further resistance.

From an anatomical viewpoint, there are a couple of ways in which these mutations resist treatment. In one, the target itself is changed — that is, the mutations change the structure of the bacterial protein



ISTOCK

near which Rifampicin normally attaches itself. This prevents the drug from binding with and killing the bacteria.

Two, some non-RRDR mutations occur alongside typical mutations, helping the bacteria recover normal growth and vigour.

DETECTION TOOLS

The authors of the paper urge health systems to move beyond testing the narrow RRDR section and to adopt newer and more comprehensive tools.

The emerging technologies that can help detect the entire set of res-

istant mutations include whole genome sequencing (WGS) and targeted next-generation sequencing (tNGS).

Tools like Deeplex Myc-TB have already demonstrated the detection of the V170F and I491F mutations from sputum and stool samples, the authors point out.

In labs in rural areas, where expensive sequencing is unavailable, options such as multiplex allele-specific PCR (MAS-PCR) and high-resolution melting analysis (HRMA) offer faster, cheaper alternatives.

Nanopore sequencing is an upcoming technology — the portable, real-time sequencing technology can detect non-RRDR mutations and may be suited for local-level testing.

For countries with high TB burdens, such as India, moving to diagnostic tools to detect non-RRDR mutations is vital in halting the silent spread of untreatable TB strains.

Logistics infrastructure creation over the past 12 years



Sector	Built up to 2014	Built after 2014 (up to 2026)	% of overall capacity built
Dedicated freight corridors	0 km	2,843 km	100
Inland waterways cargo	29 MMT	218 MMT	87
Expressway network	1,000 km	6,700 km	85
National waterways	5	32	84
Metro rail	250 km	1,000 km	75
Ports capacity	800 MTPA	2,760 MTPA	69
Railway electrification	21,800 km	69,800 km	69
4-lane & above NH	18,371 km	45,516 km	60
Airports	74	164	55

Source: PIB

GLOBAL BYTES.

Emirates, SA Airways expand tie-up

Emirates and South African Airways (SAA) have announced an expansion of their codeshare partnership, enabling access to nine cities and the start of a reciprocal codeshare agreement between the carriers. The expanded partnership opens more destinations across Central and Southern Africa. Since launching their codeshare agreement, Emirates and SAA have operated on a unilateral basis, with SAA placing its code on Emirates-operated flights between



Dubai and Johannesburg, Cape Town and Durban, as well as interline access to 68 Emirates destinations worldwide via Dubai. With regulatory approvals secured, the new agreement will see Emirates place its code on SAA-operated flights.

Joining hands to keep runways safe

Three global aviation bodies — International Air Transport Association, Civil Air Navigation Services Organisation and Airports Council International — have launched a joint initiative to reduce the risk of runway incursions. A runway incursion is defined as an aircraft, vehicle or person being incorrectly present on a runway or within its protected area. The Joint Runway Safety Initiative focuses on three areas — improving safety reporting and data capture, strengthening local runway safety teams and sharing progress and lessons learned.



More women join sailing workforce

Mitsui OSK Lines (MOL) Ltd has announced that the MOL Magsaysay Maritime Academy (MMDA), a merchant maritime academy jointly operated by MOL and Magsaysay Maritime Corporation in Manila, the Philippines, has enrolled 208 cadets, including 31 women, for the academic year 2026. The institute recorded its highest-ever enrolment of women, bringing the cumulative number of women cadets admitted to MMDA since its establishment to more than 100. This milestone reflects MMDA's commitment to promoting diversity and inclusion in maritime education and providing opportunities for talented young people regardless of gender in a field that has traditionally been male-dominated, says a company release.



RE POSTS.

Green mobility and industrial innovation are steering India-Norway trade towards new opportunities. From ships and electric vehicles to agri-based products and industrial supplies, growing bilateral trade reflects a shared commitment to sustainable growth, clean transportation, and resilient supply chains.

Ministry of Ports, Shipping and Waterways
@shipmin_india

Your opportunity to be part of India's aviation growth begins here! Applications for 389 Manager & Junior Executive posts at the Airports Authority of India open on 8 August 2026 across multiple disciplines. Mark your calendar- applications open on 8 August 2026. For more details, visit <http://aaai.aero>

Airports Authority of India
@AAI_Official

Truck rental rates

Rentals for 18-tonne payload

Truck route	Rentals as on		% change
	Jul 23	Aug 6	
Delhi-Mumbai-Delhi	1,75,000	1,74,000	-0.57
Delhi-Nagpur-Delhi	1,60,000	1,62,000	1.23
Delhi-Kolkata-Delhi	1,79,000	1,80,000	0.56
Delhi-Guwahati-Delhi	1,97,000	1,96,000	-0.51
Delhi-Hyderabad-Delhi	2,04,000	2,02,000	-0.99
Delhi-Chennai-Delhi	2,20,000	2,20,000	0.00
Delhi-Bengaluru-Delhi	2,26,000	2,28,000	0.88
Delhi-Ranchi-Delhi	1,70,000	1,72,000	1.16
Delhi-Raipur-Delhi	1,65,000	1,67,000	1.20
Delhi-Kandla-Delhi	1,44,000	1,45,000	0.69
Mumbai-Chennai-Mumbai	1,73,000	1,74,000	0.57
Ludhiana-Hyderabad-Ludhiana	2,01,000	2,03,000	0.99
Mumbai-Kolkata-Mumbai	2,03,000	2,05,000	0.98
Chennai-Ahmedabad-Chennai	2,14,000	2,15,000	0.47
Bengaluru-Kolkata-Bengaluru	2,15,000	2,17,000	0.92
Bengaluru-Mumbai-Bengaluru	1,47,000	1,48,000	0.68
Delhi-Thiruvananthapuram-Delhi	2,85,000	2,86,000	0.35
Guwahati-Mumbai-Guwahati	2,85,000	2,87,000	0.70
Nagpur-Chennai-Nagpur	1,67,000	1,69,000	1.18
Kolkata-Guwahati-Kolkata	1,60,000	1,62,000	1.23
Indore-Delhi-Indore	1,48,000	1,50,000	1.33
Ahmedabad-Indore-Ahmedabad	1,05,000	1,03,000	-1.94
Vijayawada-Mumbai-Vijayawada	1,65,000	1,67,000	1.20
Vijayawada-Kolkata-Vijayawada	1,56,000	1,55,000	-0.65
Mumbai-Nagpur-Mumbai	1,17,000	1,18,000	0.85
Mumbai-Pune-Mumbai	72,000	74,000	2.70
Mumbai-Aurangabad-Mumbai	85,000	87,000	2.30
Mumbai-Nashik-Mumbai	70,000	72,000	2.78

Figures in ₹

Source: Subham Freight Carriers India Pvt Ltd

● MARITIME INVESTMENT

Gujarat's port reforms at crossroads

LOOMING DEADLINE. Thirty years after the State shook hands with private players, its coastal economy faces a crucial test

Avinash Nair

When Gujarat opened its coastline to private investors in the late 1990s, it had a straightforward proposition: Build world-class ports, operate them for 30 years and help transform the State into India's maritime gateway. Nearly three decades later, that proposition is approaching its first real test.

The operating rights for Gujarat's first generation of privately developed ports — Pipavav and Mundra — are entering the final stretch of their concession period, putting the State's next phase of port policy under the spotlight. With newer maritime projects in States such as Andhra Pradesh, Odisha and Kerala moving towards longer concessions, Gujarat's approach to future port agreements could shape the next cycle of private investments along its coastline.

The first major private port concession agreement set to expire in Gujarat belongs to APM Terminals Pipavav, part of the Netherlands-based APM Terminals group. The port's 30-year build-own-operate-transfer, or BOOT, concession agreement was signed in 1998 and is valid till September 29, 2028.

Next up is Mundra port, operated by Adani Ports and Special Economic Zone (APSEZ), with the concession expiring on February 16, 2031. Collectively, these assets have attracted billions of dollars in investments and helped Gujarat emerge as India's largest maritime State, handling a substantial share of the country's cargo traffic.

IN A LIMBO

However, despite the looming deadlines, the State government is yet to announce a policy framework for extending the operating rights for the existing ports. According to the Gujarat Maritime Board (GMB), APM Terminals Pipavav has twice sought an extension of its concession for Pipavav port — in 2011 and 2021. APSEZ submitted similar applications for Mundra port in 2015 and 2021.

The GMB acknowledged that discussions have been underway for several years now. "The Gujarat government had sought the approval of the Union government in this regard. However, a formal announcement is yet to be made," a senior GMB official told *businessline*.

● HUB IN THE MAKING

Kerala's new maritime goals



PAUL ANTONY

The announcement that Kerala will pursue a port-led growth strategy is a welcome step, but its success will depend on the creation of a comprehensive maritime ecosystem rather than merely expanding port infrastructure for cargo loading and unloading.

Singapore offers the best model for Kerala to emulate. Beyond cargo handling, it has built a global maritime hub around bunkering or fuelling of ships (around 45 million tonnes), freshwater supply, supply of ship stores (victualling), waste collection, ship repair, supply of spare parts, crew changes and other marine services. These activities generate enormous employment and are supported by a seamless regulatory environment and a large fleet of barges providing ship-to-ship services. Supplying ships through barges on the seas saves the ships the time and effort needed to travel to the harbour, including port clearances and other formalities.

Cumbersome customs and immigration procedures in India, on the other hand, reduce our agility significantly. India imposes 5 per cent GST on bunkering, which forces our oil marketing companies to sell fuel oil in bulk to Singapore, which then does the higher-value retail bunkering to ships.

Kochi is well-placed to emulate the Singapore model. Access to high-quality bunker fuel from Kochi Refinery and world-class repair facilities at Cochin Shipyard give it the basic ingredients for a maritime services hub.

Similar capabilities can be developed at Vizhinjam through the public-private partnership model.

These measures, however, require strong support from the Central government through reforms in taxation, procedures and maritime regulations. Encouragingly, the Maritime Amrit Kaal Vision 2047 identifies the Cochin-Vizhinjam port cluster as a strategic national hub, providing the



BIZ BEYOND CARGO. Vizhinjam transshipment container port BLOOMBERG

policy foundation for coordinated action between the Centre and Kerala. The Kerala government invested around ₹6,000 crore in the Vizhinjam project, but it cannot afford to fund more such projects.

The Gujarat Maritime Board has developed Mundra, Pipavav, Hazira and Dahej ports through a model where the concessionaire created the port infrastructure. GMB earns around ₹1,700 crore a year from these ventures. Kerala must follow the Gujarat model for future port projects.

VIABILITY

However, Gujarat has the advantage of tapping the vast north Indian hinterland for cargo. Future port projects in Kerala will have no such benefit. Moreover, coastal navigation becomes difficult during the monsoon. Given these constraints, there may be limited potential for major investments in newer ports on Kerala's coastline.

On the other hand, there is significant potential in developing inland waterways in Kerala for freight, passenger traffic and tourism. For instance, it is estimated that 180-220 tanker trucks transport petroleum products daily from Kochi to Thiruvananthapuram, while 120-160 more trucks operate on the Kochi-Kozhikode route.

Diverting a substantial portion of this traffic to waterways would reduce congestion on the National Highways, lower pollution and improve road safety.

The usage of National Waterway-3 needs to be promoted, begin-



MULTICRORE ASSET. Adani Ports and Special Economic Zone's right to operate Mundra port ends on February 16, 2031 BLOOMBERG

The absence of clarity is increasingly viewed as an investment issue rather than merely a contractual one.

Port infrastructure requires continuous investments in dredging, mechanisation, deepening of berths, rail connectivity and cargo handling facilities. Without knowing whether they will continue to operate beyond the original concession period, developers could become more cautious about committing fresh capital entailing long payback periods.

This concern surfaced publicly in October 2025, when APM Terminals Pipavav signed a non-binding ₹17,000-crore investment memorandum with the Gujarat government, but indicated that major investment decisions would await clarity on the concession period.

APSEZ also indicated that the industry is closely watching how the Pipavav case unfolds. After announcing the company's first-quarter FY27 results recently, Chief Executive Officer Ashwani Gupta told investors that the company expects the Pipavav decision to precede Mundra's.

He added that discussions were underway and the company would announce developments once they became official. On whether Mundra could receive a similar treatment, he said it was too early to comment but stressed that preserving the port's scale, efficiency and competitiveness would remain important.

Gujarat was among the first States to invite private investment in port development. Its 1997 BOOT framework gave private developers the right to build and operate ports for 30 years, after which the assets would revert to the government unless the concession agreement was extended.

EVOLVING SECTOR

Meanwhile, as private participation in the maritime sector evolved, newer port projects in other States offered longer concession periods.

In Andhra Pradesh, ports such as Gangavaram and Krishnapatnam were structured with 30-year concessions and provisions for a 20-year extension. Both ports are now part of the APSEZ's portfolio after the Adani Group acquired controlling stakes.

Kerala's Vizhinjam International Deepwater Multipurpose Seaport Project was structured with an initial concession period of 40 years, and a 20-year extension under specified conditions.

Odisha, too, adopted longer concession structures for private port development. The Dhamra port concession agreement is valid for 34 years initially, including the construction period, and provides for an extension of the operating period.

A new shipbuilding policy unveiled recently by the Gujarat government may

have offered the clearest indication yet of the direction the State could take. While the policy is aimed at promoting shipbuilding, one provision caught the attention of the maritime industry. Developers setting up new shipyards can now secure waterfront concessions of up to 50 years, replacing the older 30-year tenure.

Industry executives say the move reflects the State's realisation that large maritime infrastructure requires long-term policy certainty to justify multibillion-rupee investments.

For the six greenfield ports proposed along the State's coastline, the GMB indicated that the concession period could be 30-50 years.

BALANCING INTERESTS

The proposed ports — Nana Layja (Kutch), Vadhera (Amreli), Vadodra Jhala (Gir Somnath), Damka (Surat), Lakhanka (Bhavnagar) and Bhogat (Devbhumi Dwarka) — will be developed under the BOOT model. Comprising 3,200 hectares in all, these projects are expected to expand Gujarat's cargo-handling capacity and attract an initial investment of around ₹10,000 crore.

How Gujarat balances investor certainty with public ownership of strategic coastal assets will shape the future of maritime infrastructure development not just in the State but across the country too.

● SPACE FOR GROWTH

Logistics and industrial leasing on a high

BL Chennai Bureau

India's logistics and industrial real estate sector maintained its growth momentum in the first half of 2026, supported by robust occupancy demand across the logistics, manufacturing and automotive sectors. According to Cushman & Wakefield's H1 2026 Logistics & Industrial MarketBeat report, gross leasing volume across the top eight cities reached 36.2 million sq ft (MSF), the highest ever in the first half of a calendar year.

Leasing activity grew 18 per cent year-on-year (y-o-y) and was 2 per cent higher than in H2 2025, underscoring the sector's resilience amid an evolving global business environment.

Warehouse leasing continued to dominate the market, accounting for 24.3 MSF of leasing volume and 67 per cent of total absorption. The segment recorded 11 per cent y-o-y growth, although activity moderated marginally by 6 per cent compared to H2 2025.

Industrial leasing reached nearly 12 MSF, up 36 per cent y-o-y and 27 per cent higher than in H2 2025, highlighting the rising role of manufacturing-led expansion in India's logistics and industrial real estate market, says the report.

SECTORAL TRENDS

Together, third-party logistics and engineering and manufacturing accounted for more than 60 per cent of overall leasing activity, underscoring the enduring strength of India's manufacturing and supply chain ecosystem.

The automobile sector emerged as a key growth driver, accounting for 4.8 MSF of leasing volume and 13 per cent of overall demand.

Leasing by automobile players nearly doubled compared to H1 2025, with industrial facilities accounting for nearly 63 per cent, supported by accelerated investments across the electric vehicle (EV) value chain, the report said.

Among cities, Delhi NCR emerged as the largest logistics and industrial market, followed by Chennai and Pune. Abhishek Bhutani, Managing Director, Logistics and Industrial Services, Cushman & Wakefield, said, "India's logistics and industrial sector has built on the strong momentum of 2025 to deliver a record first half in 2026, reinforcing the structural strength of occupier demand."

City-wise growth in industrial leasing demand (in million sq ft)

City	H1 '26	H1 '25	Growth (%)
Ahmedabad	22	1.7	31
Bengaluru	4.4	3.3	33
Chennai	6.1	4.4	39
Delhi NCR	8.7	5.1	69
Hyderabad	2.1	2.7	-21
Kolkata	2.1	2	6
Mumbai	4.7	7	-33
Pune	5.9	4.5	32
Total	36.2	30.7	18

Source: Cushman & Wakefield

MY FIVE.

KASHIISH A NENWANI
Director, Shivtek Spechemi Industries Ltd



Power flow yoga, detox rituals, and the art of letting go

- 1 Power flow yoga:** It isn't just about physical flexibility, but also high structural stability and rhythmic breathing under tension. It teaches you how to maintain composure and steady focus when pressure builds.
- 2 Morning detox:** My morning starts with a cold-pressed detox blend of celery, ginger and lemon before caffeine hits my system. This ritual physically hits the reset button and clears brain fog before the workday begins.
- 3 Clean fuel:** I view food entirely as clean fuel rather than an emotional indulgence. My focus is on whole food, lean protein and complex carbohydrates, while strictly minimising processed ingredients. Eating clean stabilises your energy levels, eliminating the afternoon slumps, so you can maintain a high-octane operational rhythm all day.
- 4 Post-dinner walks:** No matter how demanding the day has been, a 20-minute walk post dinner is non-negotiable. It isn't just great for digestion but also acts as an active mental boundary. It's the precise window where I disconnect from screens, process the day's events and physically transition out of work mode before sleep.
- 5 Guilt-free delegation:** True leadership comes from knowing how to protect your strategic focus. I practise prioritisation — trust cross-functional teams to own operational outcomes. Stepping away from micro-management prevents corporate burnout.

CAPSULE.

Breastfeeding — a young child's powerful protector

Too many mothers and families around the world still lack access to services and interventions that enable and support breastfeeding. Yet breastfeeding is one of the simplest and most powerful ways to protect a child's health, said the World Health Organization. It provides infants and young children with essential nutrition, increases immunity, helps protect against serious illness and supports cognitive development.

For mothers, breastfeeding brings important health benefits too, helping reduce the risk of non-communicable diseases such as breast and ovarian cancers, and type 2 diabetes. Encouragingly, breastfeeding rates are rising around the world — showing that sustained investments in policies and programmes for mothers and families are working.

COMING UP.

El Nino and health risks

El Niño is a naturally occurring climate phenomenon that can alter weather patterns around the world, increasing the likelihood of extreme heat, drought, flooding, wildfires and severe storms in many regions. These events could impact health, including heat-related illness, waterborne diseases and mental health impacts. A meet on August 12 brings together WHO experts and partners to discuss the risks associated with El Niño, share knowledge on good practices and explore how governments and communities can strengthen resilience.

PUZZLING POLICY

Cancer and the insurance gap

SCANT COVERAGE. Why cancer care largely continues to be funded out of pocket by insured patients

Ankita Upadhyay

When Kanchan Pande (64) was diagnosed with breast cancer, her family believed their health insurance policy would cushion the financial blow. Instead, they found themselves paying nearly three-fourth of the treatment cost.

Kanchan said the treatment lasted over five years at a Delhi private hospital and involved chemotherapy, radiation and hospitalisation. Her husband Balaji says the first cycle of breast cancer treatment cost approximately ₹6.2 lakh, of which around ₹5 lakh was covered by insurance. While the policy was for ₹5 lakh, Kanchan needed another round of treatment costing ₹2.9–3.2 lakh.

In all, she was admitted 78 times, each stay costing about ₹24,000, over five years. “There were nine doses of chemo that we bought ourselves, costing ₹2.5 lakh per dose, but these were also not covered by the health insurance (₹5 lakh) policy,” Balaji says, adding that the policy does not cover radiation therapy. He estimates that 70–75 per cent of the treatment cost was met out of pocket by his family. “We had insurance, but it did not help as expected,” Balaji says.

Further, he alleges that the insurance company changed its policy, without informing him, from the second year. It also limited coverage to around ₹55,000 per year for the cancer treatment, and even that was not fully reimbursed, he adds. The coverage has dropped from ₹5 lakh to ₹50,000 per year for chemotherapy.

Balaji filed petitions and consulted experts to challenge the decision, but nothing came of it, he says, as apparently that was the norm. He still annually renews his ₹5 lakh health policy.

A Delhi-based insurance agent explains that when patients do not read the minute details of an insurance policy, the “ombudsman then resolves such cases, but most patients are unable to fight their case due to their poor



CARE WITH CAVEATS. Cancer insurance often falls short, forcing patients into disputes over the fine print ISTOCK

health”. The health insurance company remains unnamed on the request of the patient’s family, which still holds the policy.

The couple’s financial strain was compounded by the absence of a pension. Kanchan says they relied on their savings to continue the treatment.

POLICY EXCLUSIONS

Doctors say the Pandes’ experience reflects a broader problem faced by cancer patients.

Even for insured patients, policy exclusions, reimbursement caps, costly next-generation medicines and delays in claim settlement can translate into significant out-of-pocket expenditure at a time when families are already grappling with a life-threatening illness.

Dr Atul Sharma, Vice-Chairman, Medical Oncology, Max Super Speciality Hospital, Saket, says insurance coverage for cancer treatment varies widely across companies, with patients frequently forced into disputes over policy exclusions and technical interpretations.

“Every insurance company has a different policy. Often patients

face challenges because insurers look for loopholes to deny or limit claims,” he alleges.

He observes that insurers sometimes question claims on the basis of a patient’s family history of cancer or classify newer therapies such as targeted therapy, oral chemotherapy and immunotherapy differently under policy terms, resulting in disputes or delays. Another common hurdle, he says, is that patients are often asked to pay for the treatment first and then seek reimbursement later, instead of being granted cashless facility.

Calling for reforms, Dr Sharma says insurers should adopt clearer guidelines and a patient-centric approach. “Patients need a more humane approach. Insurance should support them during serious illnesses rather than forcing them to fight over technicalities and the fine print,” he says.

Dr Abhik Chattopadhyay of Axiomax Oncology Private Limited, Kolkata, says the gap between insurance coverage and the actual cost of cancer treatment has widened as oncology has shifted towards expensive, targeted therapies and immunotherapy.

He says cancer patients generally depend either on government-funded schemes such as Ayushman Bharat PM-JAY and State health insurance programmes or on private insurance, but all of them have limitations.

While PM-JAY provides health cover up to ₹5 lakh annually, Chattopadhyay says the amount is inadequate for advanced cancer treatment. “Many next-generation therapies cost ₹1.3–1.5 lakh per dose,” he points out. He cites drugs such as Osimertinib for lung cancer and immunotherapy medicines such as Pembrolizumab and Nivolumab, which have significantly improved survival rates but remain unaffordable to many despite insurance.

Chattopadhyay adds that oral targeted therapies are often not covered because they do not require hospitalisation, while private insurers impose sub-limits on room rent and other expenses, reducing the final reimbursement. Expressing concern over the high cost of cancer medicines, he says the dispute over prices between insurers and hospitals leaves patients shouldering the burden of the treatment cost.

FROM THE REGULATOR

Preventing accidental overdosing on chewable pet tablets



ISTOCK

The US Food and Drug Administration is seeing a rising number of reports of pets accidentally overdosing on Apoquel Chewable, a flavoured chewable medicine approved only for dogs.

Accidental overdoses have been reported in both dogs and cats, but cats are more likely to experience serious side effects. As of April this year, there have been 259 reports of overdose in dogs and 64 in cats since the medication was approved by the FDA in June 2023.

Apoquel Chewable contains the active ingredient oclacitinib and requires a prescription from a veterinarian. It is used to control itching in dogs with skin allergies and to treat a chronic skin condition called atopic dermatitis, which is similar to eczema in people. It is safe and effective for dogs older than one year, when used at the recommended dose, but is not approved for cats.

The pork-liver flavouring in Apoquel Chewable is meant to make the medication appetising and easier to deliver to dogs. However, the flavouring also ends up tempting both dogs and cats to overfeed on the tablets, sometimes an entire bottle’s contents. An overdose of Apoquel Chewable can be life-threatening for both dogs and cats, but cats tend to develop more severe signs.

Signs of overdose in dogs and cats include vomiting and diarrhoea, lethargy (low energy or decreased activity), redness of the eyes and swelling around the eyes, and effects on the heart, brain, kidneys and liver. These signs usually appear within 24 hours after an overdose. The FDA encourages consumers to report untoward incidents to Zoetis, the manufacturer of Apoquel Chewable.

Source: USFDA

LENS ON HEALTH



CLIPPED WINGS. A shop owner holds a bird flu flyer at a farm supplies store in Pongarua, New Zealand REUTERS

Why Indians are ageing faster than they realise

BY INVITATION



NAVIN GNANASEKARAN

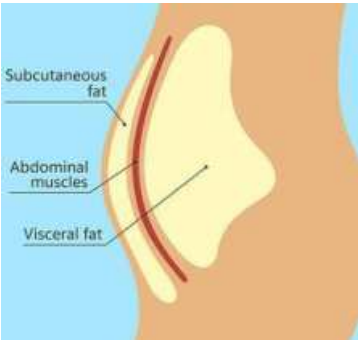
Last month, a 34-year-old software engineer walked into my clinic for what he thought was a routine scan. He was lean, vegetarian and hadn’t touched alcohol in years. His reports told an entirely different story: Early arterial stiffening, visceral fat wrapped around his organs and inflammatory markers that belong to someone two decades older. He looked at me, confused. “But, doctor,” he said, “I really thought I was healthy.”

I hear versions of this every single week. India is in the middle of a major health crisis, one that is not entirely seen or heard on the outside. We are not collapsing. We are not visibly unwell. We are, by most appearances, totally fine until such time that we aren’t. A heart attack at 48. A diabetes diagnosis at 39. A fatty liver in someone who doesn’t drink. These are not outliers anymore. They are a new pattern of urban Indian life, and it is arriving quietly, without warning, in people who genuinely believed they were doing everything right.

Non-communicable diseases are now responsible for 63 per cent of all deaths in India, claiming more than 6 million lives every single year. The problem in tackling this crisis is not a lack of effort — it is that most of what Indians know about staying healthy was not written for the Indian body.

For decades, the global framework for preventive medicine has been built on research conducted largely on Western populations, bodies that store fat differently, metabolise carbohydrates differently and carry a different genetic baseline for disease risk. We, in India, applied Western BMI cut-offs to a population that develops metabolic disease at lower body weights. We followed Western dietary proportions for a cuisine built almost entirely around carbohydrate. We used Western risk calculators for a body that develops cardiovascular disease nearly a decade earlier than its European counterpart.

The Indian body has specific vulnerabilities that are now well-docu-



DESI RISK. Indians accumulate visceral fat at body weights deemed normal under global standards

mented in research but have not yet made their way into everyday health awareness. We accumulate visceral fat — the dangerous kind that wraps around organs rather than sitting under the skin — at body weights that global standards classify as normal. We have a genetic predisposition to insulin resistance, which is activated and accelerated by a diet high in refined carbohydrate, low in protein and high in industrial seed oils. Add a generation of urban sedentary work, chronic sleep debt and stress levels that have no cultural outlet, and the result is a population ageing at a pace that its healthcare system was never designed to handle.

Chronic inflammation — now understood to be the common root cause of heart disease, type 2 diabetes, certain cancers and early cognitive decline — produces no symptoms in its early years. The damage simply accumulates, quietly, while life continues as normal.

The good news is most of this is reversible and preventable. We now know what slows ageing — muscle mass and strength, proportioned nutrition, cardiovascular fitness, sleep quality and quantity, metabolic health, stress management, and meaningful social connections.

Adding years to life has been one of medicine’s greatest achievements. But helping people live those extra years with purpose, strength, independence and dignity may well become its greatest responsibility.

The writer is a longevity physician, Head of Radiology at Apollo Hospitals, and the author of ‘The 100-year Blueprint’, recently published by Pan Macmillan India

Why a one-size-fits-all wellness approach is failing Gen Z

With rising chronic illness and anxiety, the youngest members of India’s workforce seek accessible, personalised care

J Mangaiyarkarasi

Twenty-four-year-old Ravi* knows that the treatment for the ligament tear in his knee could cost lakhs — more than what he can afford early in his career. While his employee health insurance covers it, what he seeks goes beyond hospital bills. Mental health support and preventive screenings, he says, echoing the needs of many young employees today.

In a recent report by MediBuddy-CII, over 90 per cent of the under-25 employees surveyed said they experienced symptoms of anxiety, yet fewer than half go for annual health check-ups.

As the Gen-Z cohort in their 20s enters India’s workforce, healthcare providers will need to find the right language to communicate with them. Young employees across income groups who spoke with



WHAT WORKS. Gen Z employees look for holistic healthcare ISTOCK

Pulse expressed their need for engaging and affordable healthcare programmes, in a language that is not complicated or intimidating.

As 28-year-old Kumaran explains, “The complex jargon in insurance forms put me off from buying one. So I stuck with the corporate insurance provided by my company, which is what I could afford too. Only when I fell sick and

needed tests did I discover the benefits that were offered and those that were not.”

His experience echoes that of many middle-class Gen Z professionals, for whom employer-provided insurance is often the first, and only interaction with formal healthcare financing.

Corporate India’s wellness programmes include health insurance, annual check-ups and

therapy sessions. But participation remains low, the MediBuddy report found.

VARYING NEEDS

The problem is with the ‘one-size-fits-all’ approach for a workforce that spans four generations, each with varying health priorities and expectations, the report added.

“The mental health app that resonates with a 24-year-old Gen Z hire feels clinical to a 60-year-old Baby Boomer. Most corporate wellness programmes are designed for one generation and inflicted on all four,” the report notes.

Chandan Agarwal, MediBuddy Vice-President (Consumer Business and Marketing), says, “They (early-career employees) expect a connected wellbeing ecosystem that supports every dimension of their health — physical, mental and preventive. They expect healthcare to be accessible through the year rather

than only during an illness.”

Ravi agrees that the conversation should move beyond hospitalisation. “Expanding coverage for mental health consultations and preventive care would encourage people to seek help earlier,” he says.

Twenty-five-year-old Soorya* adds, “Broader coverage for managing chronic diseases, including regular screenings, medication and follow-up care, could help prevent complications.” The challenge is in designing products that are affordable and encourage healthy behaviour through wellness incentives, he says.

As lifestyle diseases like diabetes, hypertension and cardiac problems are increasingly reported among younger people, companies and insurers will need to rethink how they connect with young employees, long before they need to make a claim.

*Names changed on request

Air India pilot’s confirmatory dope test results awaited

MISHAP AVERTED. Serious incident involving a Phuket-Delhi flight being probed by AAIB

Rohit Vaid
New Delhi

The confirmatory dope test results of the pilot-in-command of an Air India Phuket-Delhi flight that suffered a sudden loss of altitude are awaited after his initial psychoactive-substance screening indicated a result requiring further testing, the Ministry of Civil Aviation said on Sunday.

The Air India flight AI2379, operated by an Airbus A320 aircraft VT-EXO, experienced a sudden loss of altitude of approximately 300 feet during cruise on August 4 before stabilising and subsequently landing safely in Delhi.

The aircraft was carrying 137 passengers, including three infants, and eight crew members, comprising two pilots and six cabin crew.

According to the Ministry, the flight crew members underwent the prescribed psychoactive-substance screen-

UNDER THE LENS

- The screening test of the pilot-in-command indicated a result requiring confirmatory testing
- The aircraft was carrying 137 passengers, including three infants, and eight crew members



ing test as part of the standard operating procedure following the occurrence.

The screening test of the pilot-in-command indicated a result requiring confirmatory testing. The samples were subsequently sent to a designated laboratory, and the final report is awaited.

‘SERIOUS INCIDENT’

The occurrence has been classified as a ‘serious incident’ and is under investigation by the Aircraft Accident Investigation Bureau (AAIB). Pending completion of the investigation and prescribed process, both flight crew

members have been taken off the roster by the Directorate General of Civil Aviation (DGCA).

“Further action, as appropriate, will be taken based on the outcome of the investigation and confirmatory test results,” said the Ministry.

The development comes as the AAIB investigates the circumstances surrounding the sudden loss of altitude and the injuries reported on board the aircraft.

The Ministry said that the aircraft subsequently stabilised and landed safely in Delhi, while injuries were reported among a few passen-

gers and cabin crew members.

The official statement did not provide details of the substance indicated by the initial screening test or the nature of the result requiring confirmatory analysis.

POST-FLIGHT TEST

On its part, Air India said that it was aware that a post-flight screening test had been conducted on the pilots in accordance with the applicable protocols.

However, the airline said the results of the test had not been shared with it and, therefore, it was not in a position to comment on any findings. Air India said it undertakes regular drug testing of crew members in compliance with civil aviation regulations, independent of any specific flight or operational incident.

“We will continue to co-operate with the relevant authorities as required,” said an Air India spokesperson.

QSR chains witness uptick in footfalls, same store sales growth in June quarter

Meenakshi Verma Ambwani
New Delhi

Leading quick service restaurant chains posted strong topline and same store sales growth on the back of an increase in footfalls and focus on value offerings.

Players said the growth came without any significant price hikes and that they expect the positive growth momentum to sustain in the upcoming festival months. In fact, for some, the June quarter marked the highest same store sales growth.

Anant Agarwal, Vice-

Chairperson, CPRL, which runs McDonald’s stores in the northern and eastern regions, told *businessline*: “Consumers are returning to their earlier consumption patterns. Players have also pivoted in some way through value offerings. So, a combination of factors, including the positive consumer sentiment coming back, has led to a correction in sales this. We are seeing good same store sales growth in the ongoing quarter.”

After witnessing pent-up demand in post pandemic year, QSR chains were witnessing challenges in the

past 2-3 years due to subdued same store sales growth driven by various macro-economic factors.

Same store sales growth (SSSG) is an important financial metric for the industry that measures the change in revenue generated by a company’s existing restaurants over a specific period, compared to the exact same period a year prior.

STORE SALES GROWTH

Restaurant Brands Asia, which runs Burger King outlets in India, posted same store sales growth of 12.6 per cent for India business in the

June quarter, its highest in the past 15 quarters.

Rajeev Varman, Group CEO, Restaurant Brands Asia, said: “We have not taken any significant pricing action to reach the 12.6 per cent growth. It’s driven by traffic that we have generated. And, we have had a very good start to Q2 as well.”

Devyani International said the KFC business posted double-digit sales growth in the June quarter and positive same store sales growth. At the same time, Pizza Hut also “delivered sequentially better SSSG numbers”, it added.

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